MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL, 4-BIT BINARY FULL ADDERS WITH FAST CARRY, MONOLITHIC SILICON

This amendment forms part of MIL-M-38510/342(USAF), dated 2 March 1984, and is approved for use within the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

TITLE: Delete and substitute as printed above.

* 1.2.3, Case outline: Add "X C-2A (20-terminal, .350" x .350"), square chip carrier package"

* 1.3, Thermal resistance, delete and substitute:

"Thermal resistance, junction-to-case (θ_{JC}):
Cases E, F - - - - - - - - - - - - - - - - (See MIL-M-38510, appendix C)
Cases 2 and X- - - - - - - - - - - - - - - - 60°C/W 2/"

1.3, Junction temperature, after (T_J) add "3/".

Bottom of page, after note 1/ add:

"2/ When a thermal resistance value is included in MIL-M-38510, appendix C it shall supersede the value stated herein.

3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening condition in accordance with method 5004 of MIL-STD-883."

PAGE 2

1.4, after Supply voltage add "(V_{CC})".

1.4, Normalized fanout, delete "2/" and substitute "4/".

3.2.4, delete and substitute:

"3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request."

Bottom of page, delete note "2/" and substitute "4/".

PAGE 3

* TABLE I, I_{IL1} and I_{IL2}, Min limits: Delete "-0.5" and "-0.25" and substitute "0" and "0", respectively.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
9	Reprinted without change
9A	New page
10	Reprinted without change
11	Reprinted without change
12	12

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PAGE 4

3.6, delete second sentence.

TABLE II, Interim electrical parameters, delete "(pre burn-in)".

4.2.a(1), after D add "E or F".

PAGE 5

4.4.3b(1), after D add "E or F".

4.4.3b(3), add new entry as follows:

"(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883."

6.1, delete in its entirety.

PAGE 6

* FIGURE 1: Delete "Case 2" and substitute "Cases 2 and X".

PAGE 13

* Delete footnote 2/ and substitute:

"2/

Symbol	Min/Max limits (mA) for circuit			
	A	B	C	D
I_{IL1}	-.5/-1.2	-.06/-1.2		0/-.3
I_{IL2}	-.25/-.60	-.03/-.6		0/-.3

TABLE III, footnote 4/, delete "A = 3.0 V" and "B = 0.0 V or GND" and substitute "A = 2.5 V" and "B = 0.5 V" respectively; also delete "H >" and "L <" and substitute "H $\geq$" and "L $\leq$", respectively.

TABLE III, following footnote 4/, add: "5/ Perform subgroup 7 sequence at V_{CC} = 4.5 V and again at V_{CC} = 5.5 V."

* TABLE III, Delete "Case 2" and substitute "Cases 2 and X".

PAGE 14

* TABLE IV, add circuit D, Texas Instruments.

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Custodian:
Air Force - 17

Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

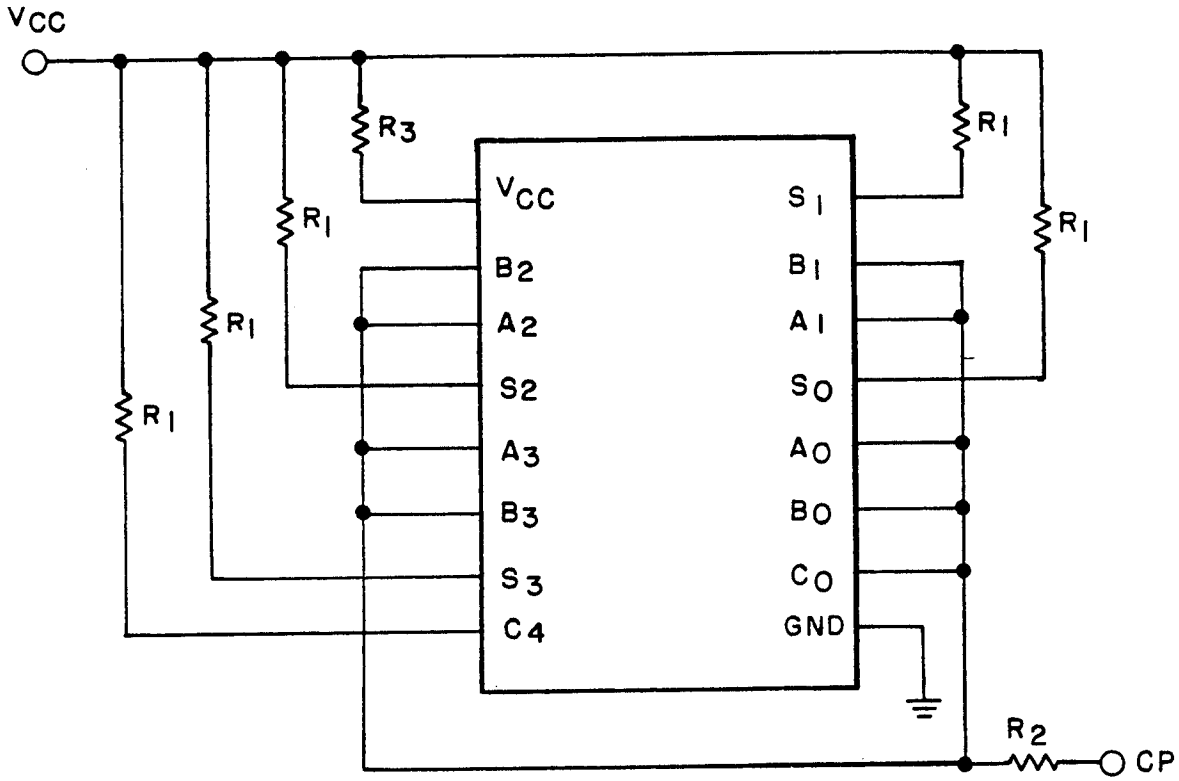
Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-F713)

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Device type 01

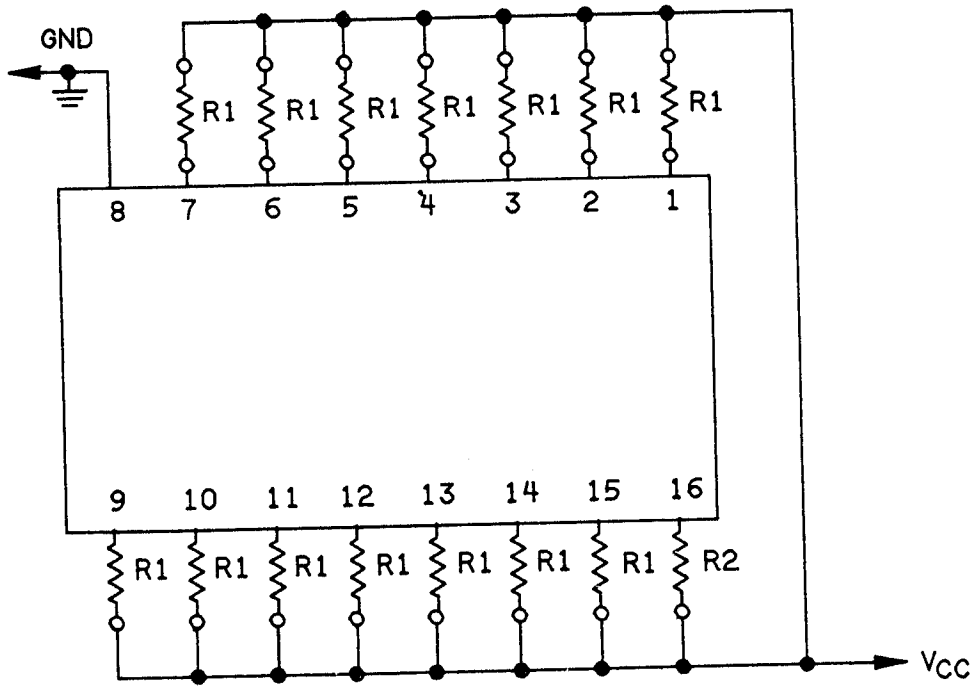


NOTES:
1. CP = 100 kHz ±50%; duty cycle = 50 ±15%; V_{IH} = 2.0 V minimum to 5.5 V maximum; V_{IL} = -0.5 V minimum to 0.8 V maximum.
2. R1 = 260Ω ±5%; R2 = 51Ω ±5%.
3. V_{CC} and R3 shall be chosen to insure a 5.5 V minimum at device V_{CC} terminal.

FIGURE 4. Burn-in and steady state life test circuit.

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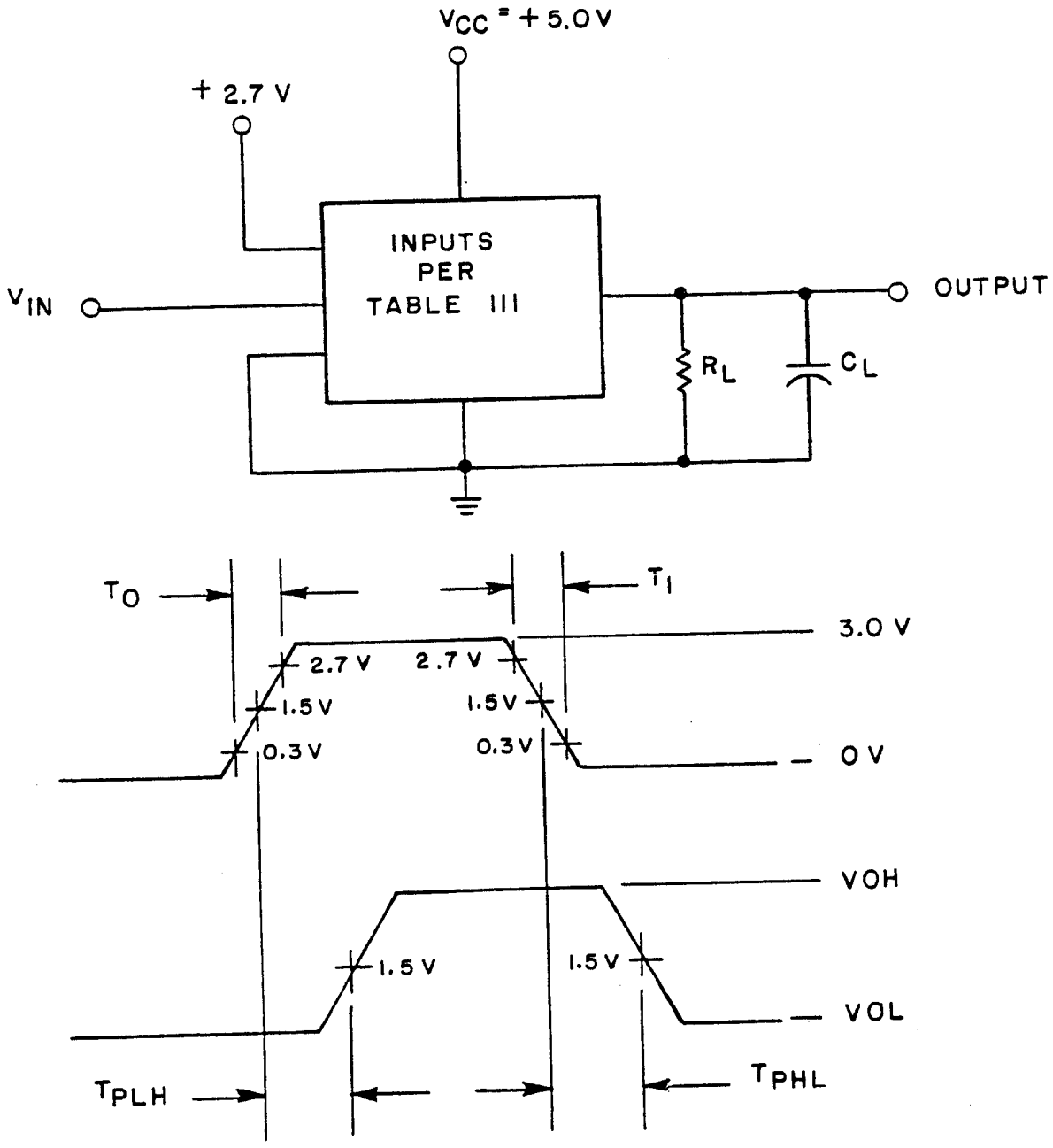
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NOTES:
1. High termperature condition F (175°C) burn-in circuit only.
2. R1 = 2 kΩ, R2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 4. Burn-in and steady state life test circuit.

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NOTES:
1. $T_0 = T_1 \leq 2.5$ ns; PRR ≤ 1 MHz, $Z_{OUT} \approx 50\Omega$.
2. $C_L = 50$ pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
3. $R_L = 499\Omega$ ±5%.
4. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Switching time waveforms for device type 01.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Cases I/2, X / Test no.	1 / 2 / S1	2 / 3 / B1	3 / 4 / A1	4 / 5 / S0	5 / 7 / A0	6 / 8 / B0	7 / 9 / C0	8 / 10 / GND	9 / 12 / C4	10 / 13 / S3	11 / 14 / B3	12 / 15 / A3	13 / 17 / S2	14 / 18 / A2	15 / 19 / B2	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	-1 mA	2.0 V	2.0 V	-1 mA	2.0 V	2.0 V	2.0 V	GND			2.0 V	2.0 V		2.0 V	2.0 V	4.5 V	SO	2.5		V
		"	2		"	"		"	"	"	"			"	"		"	"	"	S1	"		"
		"	3		"	"		"	"	"	"			"	"		"	"	"	S2	"		"
		"	4		"	"		"	"	"	"			"	"		"	"	"	S3	"		"
		"	5		"	"		"	"	"	"	-1 mA	-1 mA	"	"	-1 mA	"	"	"	C4	"		"
	V_{OL}	3007	6		0.8 V	0.8 V	20 mA	0.8 V	0.8 V	0.8 V	"			0.8 V	0.8 V		0.8 V	0.8 V	"	SO		0.5	"
		"	7	20 mA	"	"		"	"	"	"			"	"		"	"	"	S1		"	"
		"	8		"	"		"	"	"	"			"	"		"	"	"	S2		"	"
		"	9		"	"		"	"	"	"			"	"	20 mA	"	"	"	S3		"	"
		"	10		2.0 V	2.0 V		2.0 V	2.0 V	"	"	20 mA	20 mA	"	"		"	"	"	C4		"	"
	V_{IC}		11		-18 mA						"								"	B1		-1.2	"
			12			-18 mA					"								"	A1		"	"
			13					-18 mA			"								"	A0		"	"
			14						-18 mA		"								"	B0		"	"
			15							-18 mA	"								"	C0		"	"
			16								"			-18 mA					"	B3		"	"
			17								"				-18 mA				"	A3		"	"
			18								"						-18 mA		"	A2		"	"
			19								"							-18 mA	"	B2		"	"
	I_{IH1}	3010	20		GND	2.7 V					"								5.5 V	A1		20	µA
		"	21		2.7 V	GND					"								"	B1		"	"
		"	22					2.7 V	GND		"								"	A0		"	"
		"	23					GND	2.7 V		"								"	B0		"	"
		"	24							2.7 V	"			GND	2.7 V				"	A3		"	"
		"	25								"			2.7 V	GND				"	B3		"	"
		"	26								"						2.7 V	GND	"	A2		"	"
		"	27								"						GND	2.7 V	"	B2		"	"
		"	28								"								"	C0		"	"
	I_{IH2}	"	29		GND	7.0 V	7.0 V				"								"	A1		100	"
		"	30		7.0 V	GND					"								"	B1		"	"
		"	31					7.0 V	GND		"								"	A0		"	"
		"	32					GND	7.0 V		"								"	B0		"	"
		"	33							7.0 V	"			GND	7.0 V				"	A3		"	"
		"	34								"			7.0 V	GND				"	B3		"	"
		"	35								"						7.0 V	GND	"	A2		"	"
		"	36								"						GND	7.0 V	"	B2		"	"
		"	37								"								"	C0		"	"
	I_{IL1}	3009	38		5.5 V	0.5 V					"								"	A1	2/	2/	mA
		"	39		0.5 V	5.5 V					"								"	B1	"	"	"
		"	40					0.5 V	5.5 V		"								"	A0	"	"	"
		"	41					5.5 V	0.5 V		"								"	B0	"	"	"
		"	42								"			5.5 V	0.5 V				"	A3	"	"	"
		"	43								"			0.5 V	5.5 V				"	B3	"	"	"
		"	44								"						0.5 V	5.5 V	"	A2	"	"	"
		"	45								"						5.5 V	0.5 V	"	B2	"	"	"
	I_{IL2}	"	46							0.5 V	"								"	CO	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Cases 1/ 2, X / Test no.	1 / 2 / S1	2 / 3 / B1	3 / 4 / A1	4 / 5 / SO	5 / 7 / AO	6 / 8 / BO	7 / 9 / CO	8 / 10 / GND	9 / 12 / C4	10 / 13 / S3	11 / 14 / B3	12 / 15 / A3	13 / 17 / S2	14 / 18 / A2	15 / 19 / B2	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{OS}	3011	47	GND	5.5 V	5.5 V	GND	5.5 V	5.5 V	5.5 V	GND		GND	5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	S1	-60	-150	mA
		"	48		"	"		"	"	"	"			"	"		"	"	"	SO	"	"	"
		"	49		"	"		"	"	"	"			"	"		"	"	"	S3	"	"	"
		"	50		"	"		"	"	"	"			"	"		"	"	"	S2	"	"	"
		"	51		"	"		"	"	"	"	GND		"	"	GND	"	"	"	C4	"	"	"
	I_{OD}		52	2.5 V	GND	GND	2.5 V	GND	GND	GND	"								4.5 V	S1	60		"
			53							"	"								"	SO	"		"
			54							"	"								"	S3	"		"
			55							"	"		2.5 V	GND	GND		GND	GND	"	S2	"		"
			56		5.5 V	5.5 V		5.5 V	5.5 V	"	"	2.5 V		GND	GND	2.5 V	5.5 V	GND	"	C4	"		"
	I_{CC1}	3005	57		GND	GND		GND	GND	"	"			GND	GND		GND	GND	5.5 V	V_{CC}		55	"
	I_{CC2}	"	58		GND	5.5 V		5.5 V	GND	5.5 V	"			GND	5.5 V		5.5 V	GND	5.5 V	"		55	"
	I_{CC3}	"	59		5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	"			5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	"		55	"
2	Same tests, terminal condition, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C 3/	Truth table test	3014	60	L	B	B	L	B	3	B	GND	L	L	B	B	L	B	B	5.0 V 5/	All Outputs	See 4/		
			61	L	B	B	H	B	3	A	"	H	H	A	A	H	A	A	"	"	"		
			62	L	B	B	H	A	3	B	"	H	L	A	A	H	A	B	"	"	"		
			63	H	B	B	L	A	B	A	"	H	L	A	A	L	B	A	"	"	"		
			64	L	B	B	H	B	A	B	"	H	L	A	A	L	B	B	"	"	"		
			65	H	B	B	L	B	A	A	"	H	L	A	B	L	B	A	"	"	"		
			66	H	B	B	L	A	A	B	"	L	H	A	B	H	B	B	"	"	"		
			67	H	B	B	H	A	A	A	"	L	H	A	B	H	B	B	"	"	"		
			68	H	B	A	L	B	3	B	"	L	H	A	B	L	B	A	"	"	"		
			69	H	B	A	H	B	3	A	"	H	L	B	A	L	A	A	"	"	"		
			70	H	B	A	H	A	B	B	"	L	H	B	A	H	A	A	"	"	"		
			71	L	B	A	L	A	B	A	"	H	H	B	A	H	A	B	"	"	"		
			72	H	B	A	H	B	A	B	"	L	H	B	B	H	A	B	"	"	"		
			73	L	B	A	L	B	A	A	"	L	H	B	B	L	A	B	"	"	"		
			74	L	B	A	L	A	A	B	"	L	H	B	B	H	A	B	"	"	"		
			75	L	B	A	H	A	A	A	"	L	H	B	B	H	A	B	"	"	"		
			76	H	A	B	L	B	B	B	"	L	H	B	B	L	A	B	"	"	"		
			77	H	A	B	H	B	B	A	"	L	L	B	B	H	A	B	"	"	"		
			78	H	A	B	H	A	B	B	"	L	L	B	B	L	A	B	"	"	"		
			79	L	A	B	L	A	B	A	"	H	L	B	A	H	A	B	"	"	"		
			80	H	A	B	H	B	A	A	"	L	L	B	A	L	A	B	"	"	"		
			81	L	A	B	L	B	A	B	"	H	L	B	A	L	A	B	"	"	"		
			82	L	A	B	L	A	A	A	"	H	L	B	A	H	B	B	"	"	"		
			83	L	A	B	H	A	A	B	"	L	H	B	A	L	B	B	"	"	"		
			84	L	A	A	L	B	B	A	"	H	L	A	B	L	B	B	"	"	"		
			85	L	A	A	H	B	B	B	"	H	L	A	B	H	B	A	"	"	"		
			86	L	A	A	H	A	B	A	"	H	L	A	A	H	B	A	"	"	"		
			87	H	A	A	L	A	A	B	"	H	L	A	A	L	B	B	"	"	"		
			88	L	A	A	L	B	A	A	"	H	H	A	A	L	B	B	"	"	"		
			89	H	A	A	L	B	A	B	"	H	H	A	B	L	A	B	"	"	"		
			90	H	A	A	L	A	A	A	"	L	H	B	B	L	B	A	"	"	"		
			91	H	A	A	H	A	A	A	"	L	L	B	B	H	B	B	"	"	"		
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = 125°C and T_C = 55°C.																						

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